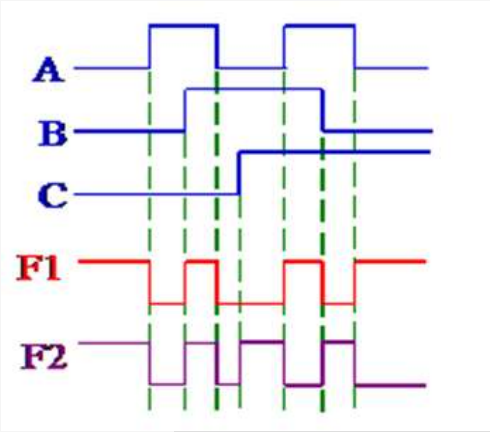
西安交通大学
XI'AN JIAOTONG UNIVERSITY

数电第三次作业

2.1

由图得：$F_1 = \overline{A \oplus B}, F_2 = F_1 \oplus C$

波形图：



2.2

$$A_1 = \overline{B_1}$$

$$A_2 = B_2$$

$$A_4 = B_2 \oplus B_4$$

$$A_8 = \overline{B_2 + B_4 + B_8}$$

结论：$B_8B_4B_2B_1$是BCD码，$A_8A_4A_2A_1$是$B_8B_4B_2B_1$对9的变补

2.3

$Z = D$

$Y = C \oplus D$

$X = B \oplus (C + Y) == B \oplus (C + D)$

$W = A \oplus (B + C + D)$

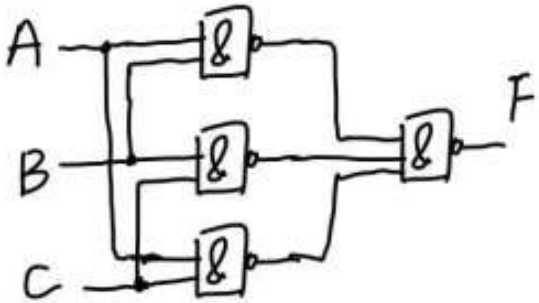
结论：16（四位二进制）变补器。

真值表为

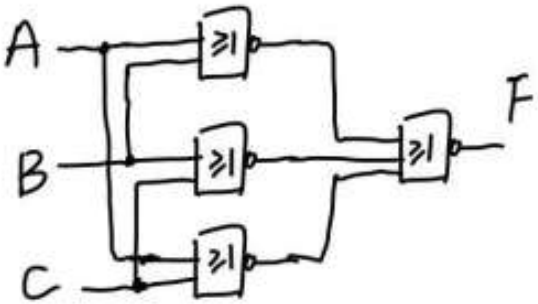
A	B	C	D	z	y	x	w
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	0	1	1	1
0	0	1	1	1	0	1	1
0	1	0	0	0	0	1	1
0	1	0	1	1	1	0	1
0	1	1	0	0	1	0	1
0	1	1	1	1	0	0	1
1	0	0	0	0	0	0	1
1	0	0	1	1	1	1	0
1	0	1	0	0	1	1	0
1	0	1	1	1	0	1	0
1	1	0	0	0	0	1	0
1	1	0	1	1	1	0	0
1	1	1	0	0	1	0	0
1	1	1	1	1	0	0	0

2.4

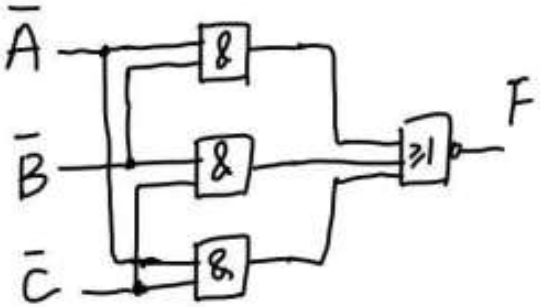
2.4 (1) 与非：$F=\overline{\overline{AB}\,\overline{AC}\,\overline{BC}}$



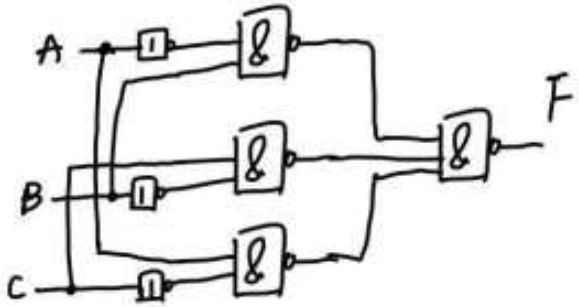
或非：$F=\overline{\overline{A+B}+\overline{A+C}+\overline{B+C}}$



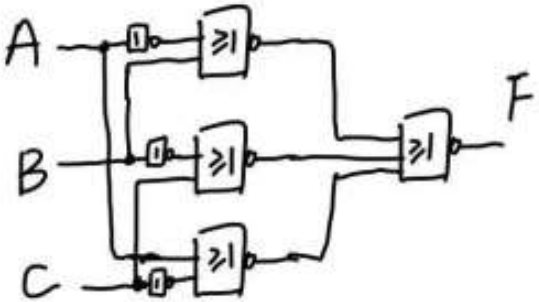
与或非：$F=\overline{A\bar{B}+\bar{A}\bar{C}+\bar{B}\bar{C}}$



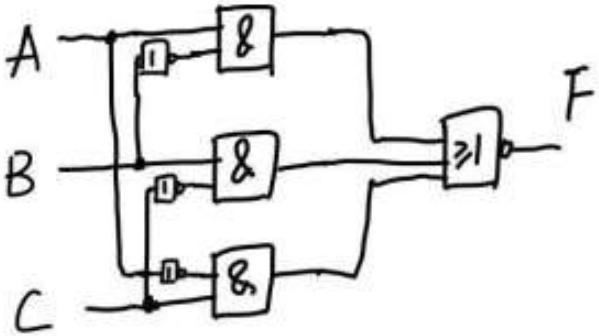
(2) 与非：$F=\overline{\overline{A\bar{B}}\cdot\overline{\bar{B}C}\cdot\overline{A\bar{C}}}$



或非：$F=\overline{\overline{A+B}+\overline{\bar{B}+C}+\overline{A+\bar{C}}}$



与或非：$F=\overline{A\bar{B}+\bar{A}C+B\bar{C}}$



2.5(2) 设输入为 ABCD

$$F=AB\bar{C}\bar{D}+A\bar{B}C\bar{D}+A\bar{B}\bar{C}D+\bar{A}BC\bar{D}+\bar{A}B\bar{C}D+\bar{A}\bar{B}CD$$

$$=m_{12}+m_{10}+m_9+m_6+m_5+m_3$$

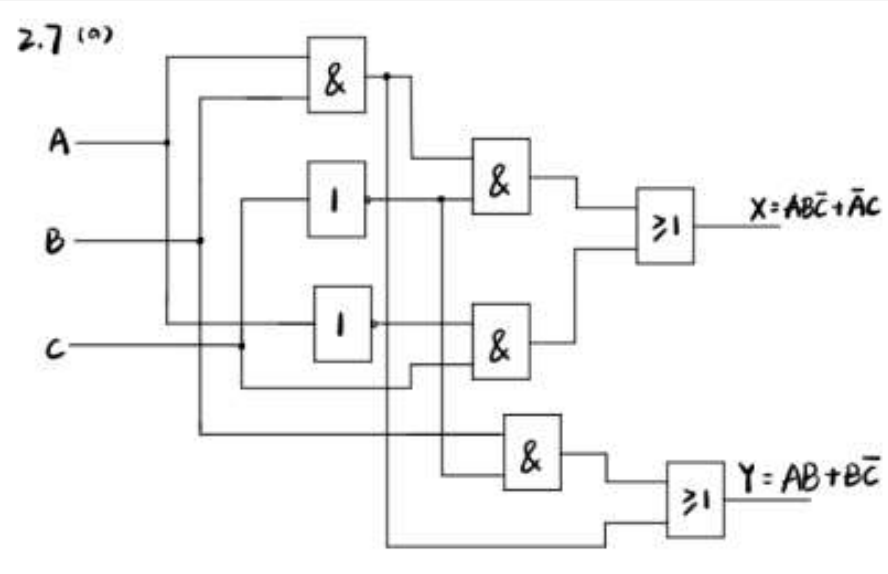
$$=\sum m^{a}(3,5,6,9,10,12)$$

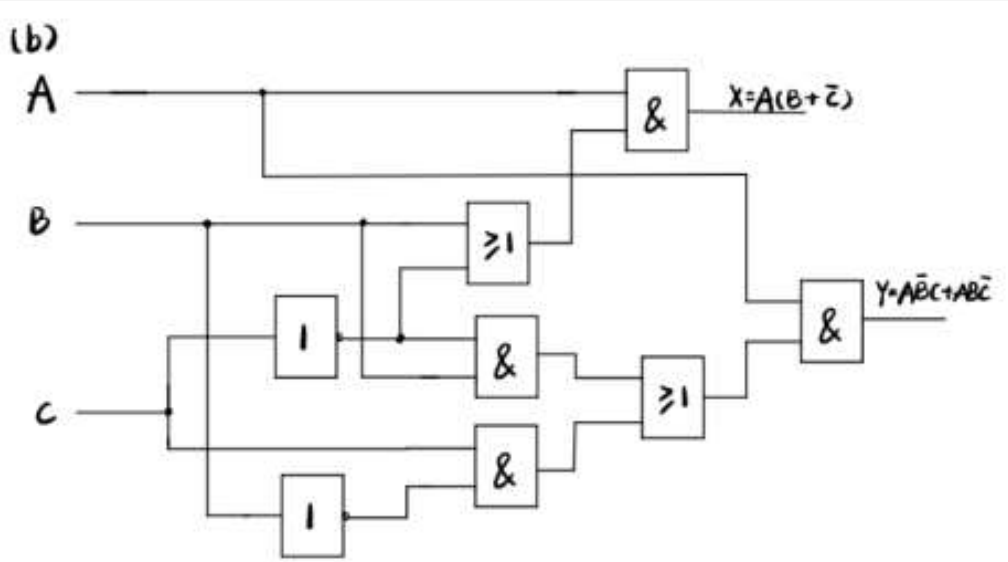
0	0	1	0
0	1	0	1
1	0	0	0
0	1	0	1

由卡诺图可知该逻辑表达式无法化简

故该电路逻辑表达式为

$$F=AB\bar{C}\bar{D}+A\bar{B}C\bar{D}+A\bar{B}\bar{C}D+\bar{A}BC\bar{D}+\bar{A}B\bar{C}D+\bar{A}\bar{B}CD$$





2.9

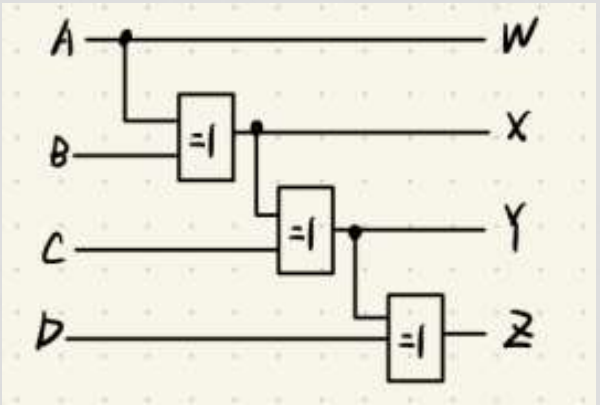
令典型格雷码为ABCD，二进制码为WXYZ，则

$W = A$

$X = A \oplus B$

$Y = A \oplus B \oplus C$

$Z = A \oplus B \oplus C \oplus D$



2.11

（1）F= $\overline{A}B + \overline{B}\,\overline{C}$+AC

BC=11时，$F = A + \overline{A}$，存在静态1险象

AC=00时，$F = B + \overline{B}$，存在静态1险象

AB=10时，$F = \overline{C} + C$，存在静态1险象

消除：添加多余项A$\overline{B}$、BC、$\overline{A}\,\overline{C}$

（2）F=(A+C+$\bar{D}$)($\bar{B} + C + D$)($\bar{B} + \bar{C}$)(B+D)

D=0时（A,C=0或1均可），$F = B \cdot \overline{B}$，存在静态0险象

AB=01时（D=0或1均可），$F = C \cdot \overline{C}$，存在静态0险象

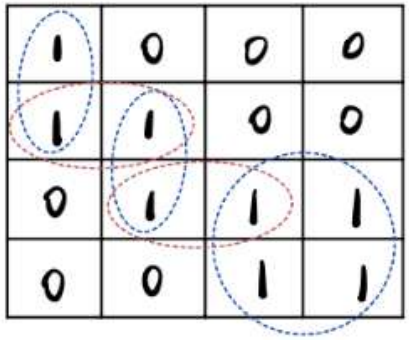
AC=00时（B=0或1都可以），$F = D \cdot \overline{D}$，存在静态0险象

消除：添加多余项（$\bar{C} + D$）、(A+$\bar{B}$+C)



2.12

2.12 (1) $F=\sum m^4(0,1,5,7,10,11,14,15)$



共有3个必要极大圈

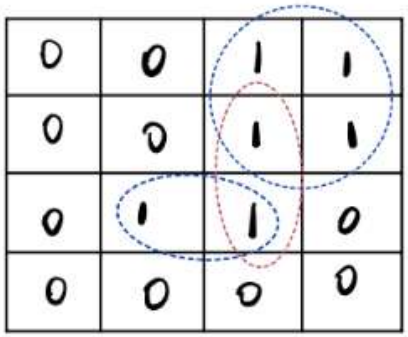
$F=\bar{A}\bar{B}\bar{C}+\bar{A}BD+AC$

存在逻辑险象，增加多余项消除

$F=\bar{A}\bar{B}\bar{C}+\bar{A}BD+AC+\bar{A}\bar{C}D+BCD$

(2) $F=\prod M^4(0,1,2,3,4,5,6,10,11,14)$

$=\sum m^4(7,8,9,12,13,15)$



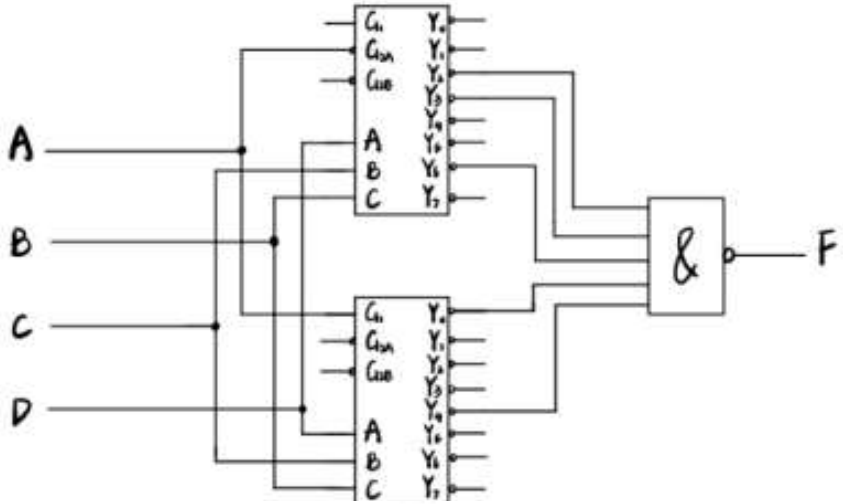
共有2个必要极大圈

$F=A\bar{C}+BCD$

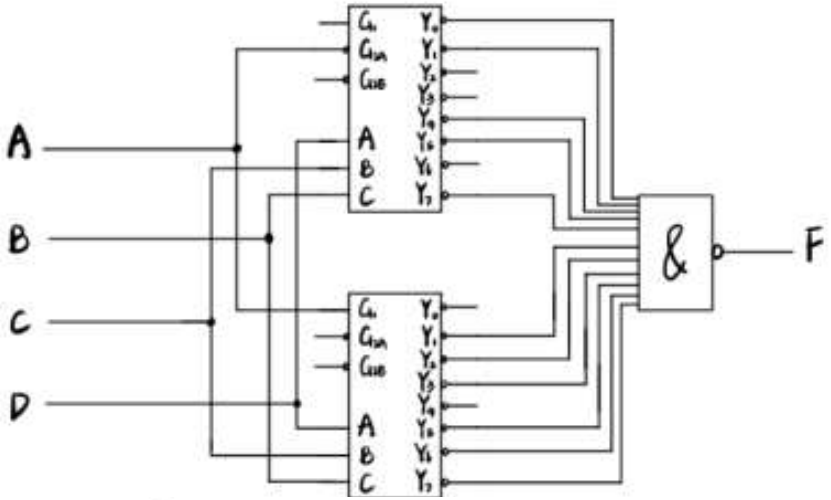
存在逻辑险象，增加多余项消除

$F=A\bar{C}+BCD+ABD$

2.13 (1) $F = \sum m^*(2,3,6,8,12)$

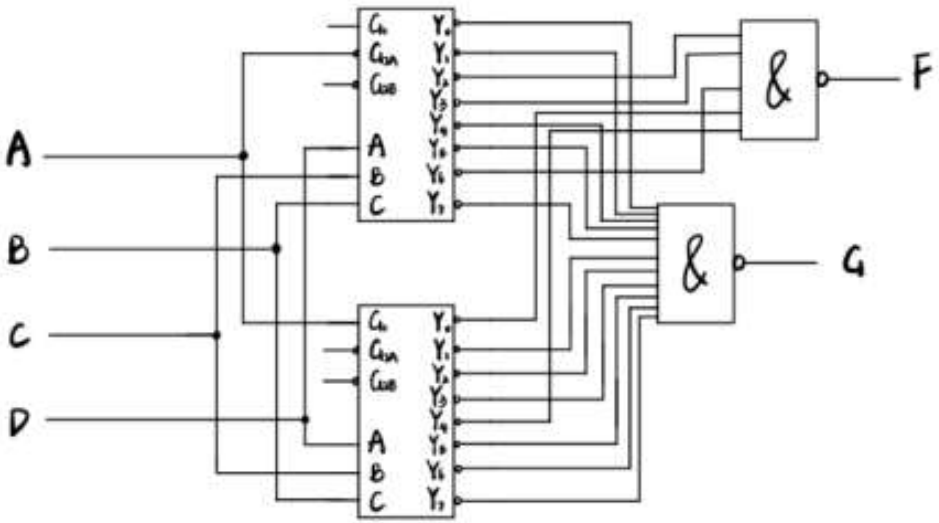


(2) $F = \prod M^*(2,3,6,8,12) = \sum m^*(0,1,4,5,7,9,10,11,13,14,15)$



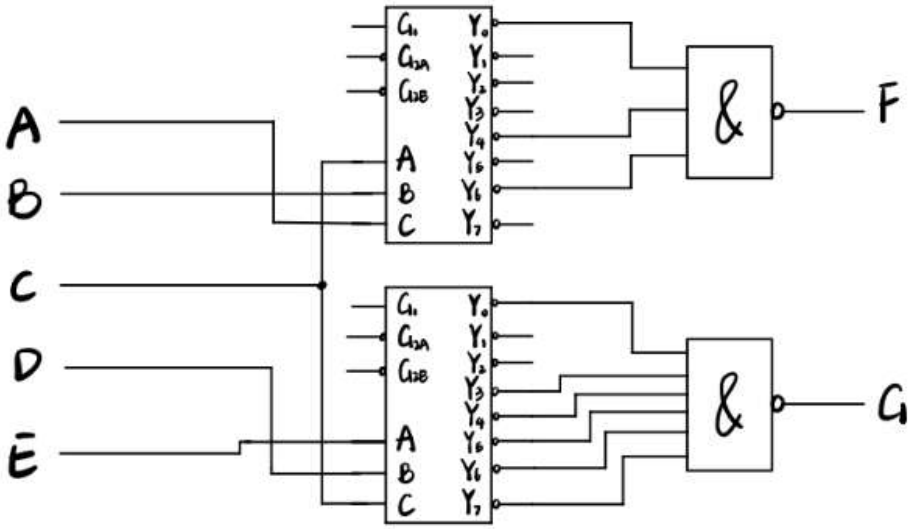
(3) $F(A,B,C,D) = \sum m^*(2,3,6,8,12)$

$G(A,B,C,D) = \prod M^*(2,3,6,8,12)$



(4) $F(A,B,C) = \sum m(0,4,6)$

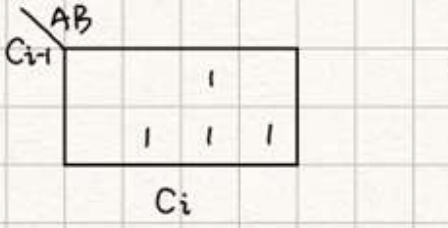
$G(C,D,E) = \prod M(1,2)$

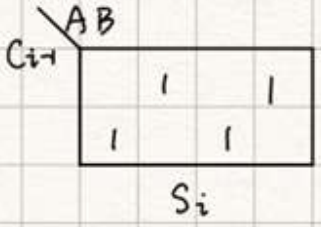


2.14

2.14 A B C_{i-1} ⇒ S_i C_i

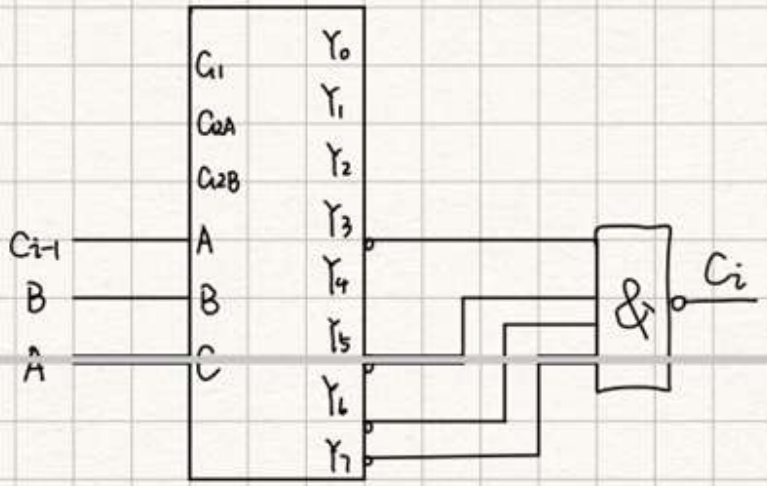
设加数为A与B，进位为C_{i-1}，结果和进位分别为S_i和C_i

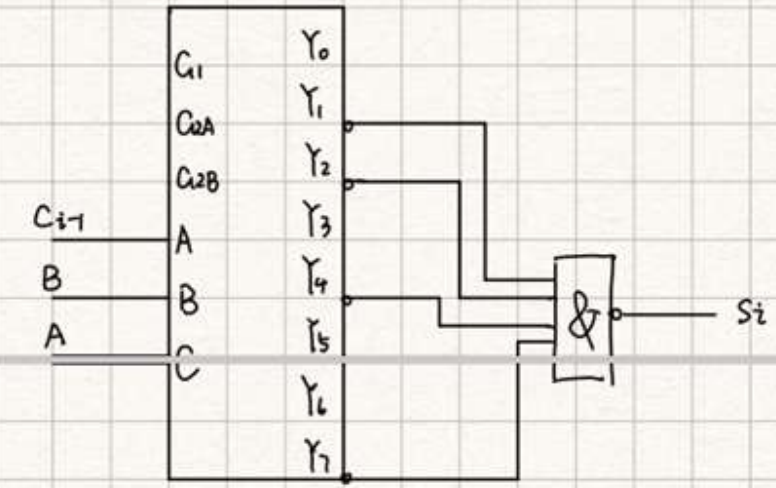




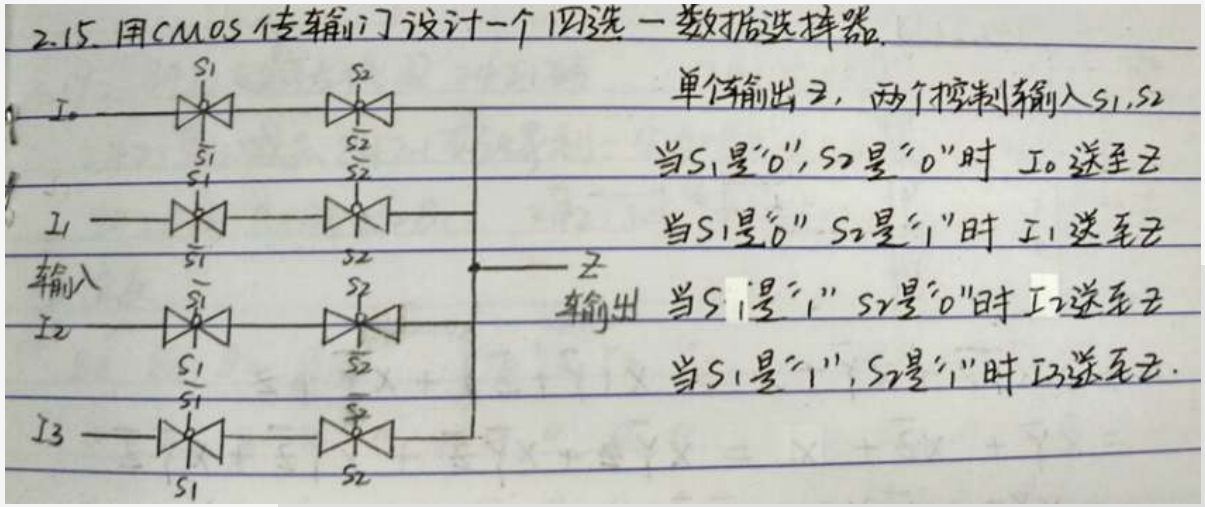
$C_i = \sum m^4(3,5,6,7)$

$S_i = \sum m^4(1,2,4,7)$

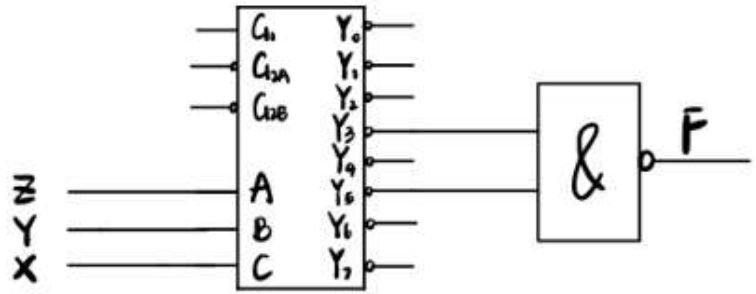




2.15


2.15. 用CMOS传输门设计一个四选一数据选择器.
单件输出Z，两个控制输入S1,S2
当S1是"0",S2是"0"时 I0送至Z
当S1是"0" S2是"1"时 I1送至Z
当S1是"1" S2是"0"时 I2送至Z
当S1是"1",S2是"1"时 I3送至Z.
输入
I0
I1
I2
I3
Z
输出
S1
S2

2.18 (1) $F = \bar{X}YZ + X\bar{Y}Z = \sum m^3(3,5)$

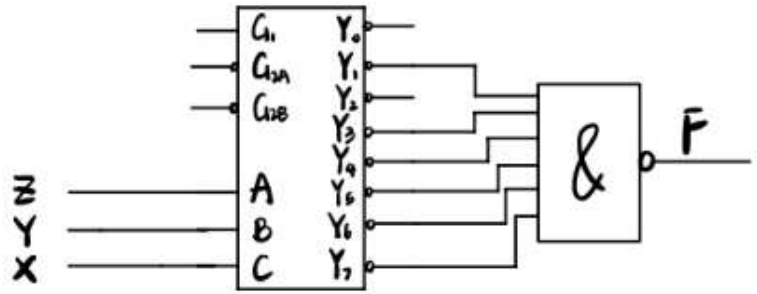


(2) $F = XY\overline{Z} + X\bar{Y} + Z$

$= X\bar{Y} + X\bar{Z} + Z$

$= X\bar{Y}Z + X\bar{Y}\bar{Z} + XY\bar{Z} + XYZ + \bar{X}YZ + \bar{X}\bar{Y}Z$

$= \sum m^3(1,3,4,5,6,7)$



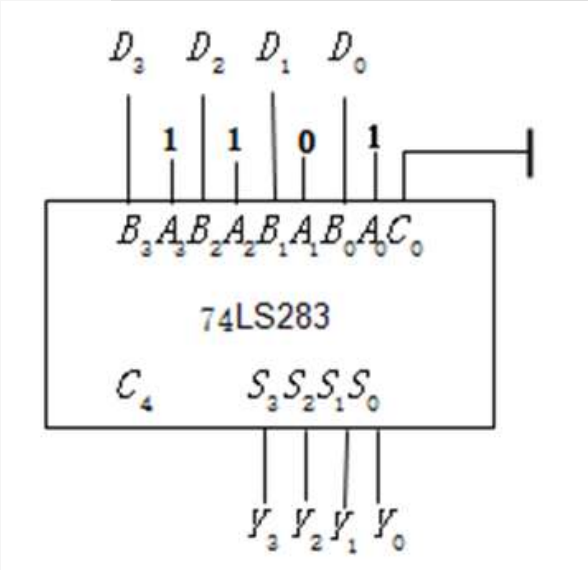
2.19

(1)用$D_3D_2D_1D_0$表示余3码, 用$Y_3Y_2Y_1Y_0$表示8421码,

则$Y_3Y_2Y_1Y_0 = D_3D_2D_1D_0 - 0011$

$= D_3D_2D_1D_0 + (-0011)$补

$= D_3D_2D_1D_0 + 1101$



2.19

(2)用$D_3D_2D_1D_0$表示余3码，用$Y_3Y_2Y_1Y_0$表示8421码

余3码的前五个数（0000-0100）可由2421码的前五个数加上3（即加上0011）得到，

余3码的后五个数（1011-1111）可由当2421码的后5个数减去3（即加上1101）得到。

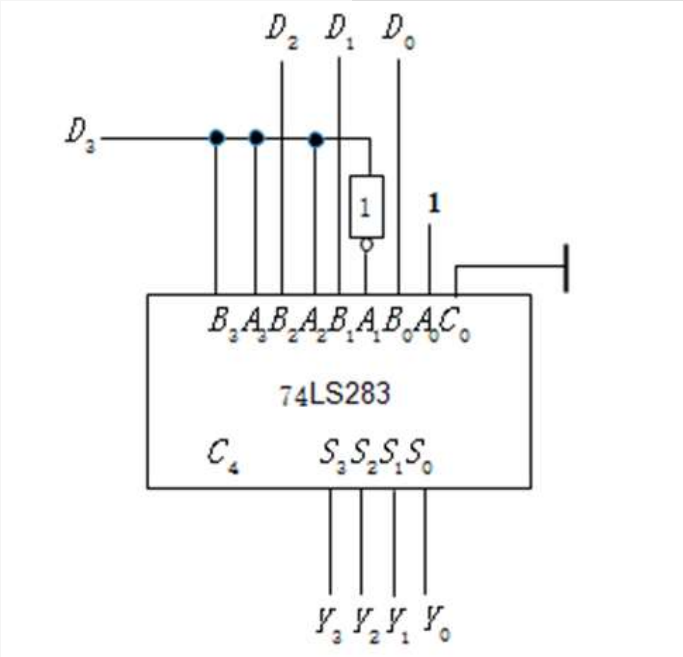
则由题可知，74LS283的输入端$A_3A_2A_1A_0$取值是变化的，

其中A_3A_2的取值与输入端B_3一致，也即与$D_3D_2D_1D_0$中的D_3一致；

输入端A_1的取值为B_3的反，也即为D_3的反；

输入端A_0取值一直为1。

则电路图如下：



注：

通过上述分析可知，

$A_3 = A_2 = B_3 = D_3$

$A_1 = \overline{B_3} = \overline{D_3}$

$A_0 = 1$

理清思路，电路图不用太过复杂